

B.Tech II Year II Semester (R20) Supplementary Examinations November 2024

EM WAVES AND TRANSMISSION LINES

(Electronics and Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

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|-----|--|----|
| (a) | Write del operator in spherical coordinate system. | 2M |
| (b) | Write about Gauss law and applications. | 2M |
| (c) | Write Maxwell's two equations for magneto static fields. | 2M |
| (d) | What is the Significance of Magnetic flux Density? | 2M |
| (e) | List the key relations between the electric field (E) and magnetic field (H) in uniform plane waves. | 2M |
| (f) | State the boundary conditions for electromagnetic fields at a dielectric-dielectric interface. | 2M |
| (g) | What is total internal reflection? | 2M |
| (h) | What is Surface Impedance? | 2M |
| (i) | What do you mean by distortion less line? | 2M |
| (j) | Write about Transmission line parameters. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|--|-----|
| 2 | (a) Define Electric field intensity and write the properties electric flux. | 5M |
| | (b) A Point charge of 20nC is located at the origin. Determine the magnitude and direction of the electric field intensity at point (1, 3, -4). | 5M |
| OR | | |
| 3 | (a) A charge of $5 \times 10^{-8} C$ is distributed uniformly on the surface of a sphere of radius 1 cm. It is a sphere of radius 6 cm. Determine electric flux density. | 5M |
| | (b) Determine the Relationship between E and V. | 5M |
| 4 | (a) Explain Ampere's Circuital Law. | 5M |
| | (b) Explain about Faraday's law and Transformer e.m.f. | 5M |
| OR | | |
| 5 | (a) Discuss about Magnetic Vector and Scalar Potentials. | 5M |
| | (b) Explain about Maxwell's equations in different final forms and word statements. | 5M |
| 6 | Explain the concept of polarization in electromagnetic fields. Discuss different mechanisms of polarization and their significance in various applications. | 10M |
| OR | | |
| 7 | Derive the wave equations for electromagnetic fields in conducting media. | 10M |
| 8 | State and Prove Poynting theorem. | 10M |
| OR | | |
| 9 | (a) Discuss in brief about power loss in a plane conductor. | 5M |
| | (b) Write about Brewster angle and Critical angle. | 5M |
| 10 | Explain the following:
(i) Smith Chart.
(ii) Stub matching. | 10M |
| OR | | |
| 11 | Explain about Transmission line equations and their solutions in their phasor form. | 10M |

B.Tech II Year II Semester (R20) Regular & Supplementary Examinations April/May 2024

EM WAVES AND TRANSMISSION LINES

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) Write about Spherical Coordinates. 2M
- (b) Write about Divergence Theorem. 2M
- (c) List the Biot-Savart's Law for all types of current distributions. 2M
- (d) Define magnetic flux density and explain its significance. 2M
- (e) State the boundary conditions for electromagnetic fields at a dielectric-dielectric interface. 2M
- (f) List the key relations between the electric field (E) and magnetic field (H) in uniform plane waves. 2M
- (g) What is Brewster angle? 2M
- (h) What is Critical angle? 2M
- (i) A low loss transmission line of 100Ω characteristics impedance is connected to a load of 200Ω . Compute the voltage reflection coefficient. 2M
- (j) A 50Ω lossless transmission line is terminated on a load impedance of $Z_L = (25 + j 50) \Omega$. Find VSWR. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Define Coulomb's law and derive the force F that exists between two unlike charges. 5M
- (b) Two-point charges, $Q_A = +8 \mu\text{C}$ and $Q_B = -5 \mu\text{C}$, are separated by a distance $r = 10 \text{ cm}$. What is the magnitude of the electric force between them? 5M

OR

- 3 (a) Apply Gauss Law to evaluate the electric flux density at a point P due to the point charge located at the origin. 5M
- (b) Define Electric Potential. Find the electric potential for a point charge is located at origin and Write Maxwell's second equation for electrostatic field. 5M

- 4 (a) Explain Biot-Savart's Law. 5M
- (b) Find the Magnetic field Intensity due to a Straight current carrying filamentary conductor of finite length. 5M

OR

- 5 (a) Determine the Magnetic Field Intensity due to a infinite sheet current. 5M
- (b) Explain about Inconsistency of Ampere's Law. 5M

- 6 Explain the boundary conditions for electromagnetic fields at the interface between two dielectric materials. 10M

OR

- 7 Explain about Wave Propagation in good conductors and good dielectrics. 10M

Contd. In Page 2

- 8 Evaluate the expressions for reflection coefficient and transmission coefficient by a normal incident wave for a dielectric medium. 10M
- OR**
- 9 Explain the following : 10M
(i) Poynting Theorem.
(ii) Power loss in a Plane Conductor.
- 10 With neat sketch explain about Primary and Secondary constants of transmission line. 10M
- OR**
- 11 Explain about Single and Double Stub Matching. 10M

B.Tech II Year II Semester (R20) Regular & Supplementary Examinations August/September 2023

EM WAVES AND TRANSMISSION LINES

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Given a vector function $F = a_x(3y - c_1z) + a_y(c_2x - 2z) - a_z(c_3y + z)$, Determine the constants c_1 , c_2 and c_3 if 'F' is irrotational. 2M
 - (b) State Gauss law. 2M
 - (c) What is the basis for Magnetic Scalar Potential? 2M
 - (d) Write the integral form of Maxwell's equations. 2M
 - (e) What is a boundary condition? How do boundary conditions arise and how are they derived? 2M
 - (f) Most microwave ovens operate at 2.45 GHz. Assume $\sigma = 1.1 \times 10^6 \text{ } \Omega/\text{m}$ and $\mu_r = 600$ for the stainless steel interior and find the depth of penetration. 2M
 - (g) What is the Poynting Vector? What is the physical interpretation of the Poynting Vector over a closed surface? 2M
 - (h) Define Brewster angle. 2M
 - (i) Draw the Voltage and Current variation along an Open circuited and a Short circuited line. 2M
 - (j) Draw an Equivalent Circuit of Two Wire Transmission Line. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Point charges Q_1 and Q_2 are respectively located at (4, 0, -3) and (2, 0, 1). If $Q_2 = 4\text{nc}$, Find Q_1 such that (i) the E at (5, 0, 6) has no Z-Component, (ii) the force on a Test charge at (5, 0, 6) has no X-Component. 10M
- OR**
- 3 Determine the Electric field intensity of an infinitely long straight, line charge of uniform density ρ_L in air. 10M
- 4 Explain the nature of line, surface and volume current distributions as applicable to static magnetic fields. List out the expressions for the magnetic field intensity in these three cases. 10M
- OR**
- 5 Find the magnetic field intensity at the center of a square loop, with side 'w' Carrying a direct current 'I'. 10M
- 6 From the Maxwell's curl's equation derive the wave equations for an Electromagnetic wave in conducting media. 10M
- OR**
- 7 Write short notes on EM wave Polarization. 10M
- 8 Discuss the determination of the reflected and wave fields of a uniform plane wave incident normally onto a plane boundary between two material media. 10M
- OR**
- 9 State and prove Poynting Theorem. 10M
- 10 Show that a Transmission line will be distortion less if $LG = RC$. 10M
- OR**
- 11 Define input impedance of a transmission line. Derive an expression for input impedance of a transmission line in terms of reflection co-efficient. 10M
